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Practice Quiz. 2.2.2. Unsupervised Anomaly-based Malware Detection Techniques

coach

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1. In "Unsupervised Anomaly-based Malware Detection using Hardware Features" presentation, the author advocated what are advantages of using Hardware Features to detect malware? **1 / 1 point**

Select all that apply.

- [x] Cheaper to collect data/features.

  Correct
  Correct

- [x] Only require minimal features to detect the presence of malware.

  Correct
  Correct.

- [x] Hard to evade by malware designer.

  Correct
  Correct.

2. In "Unsupervised Anomaly-based Malware Detection using Hardware Features" research, the main idea of the authors is to **1 / 1 point**

- (●) Detect
  anomalies caused by the perturb microarchitecture behavior of the attacked program by the malware.
- ( ) Detect
  anomalies caused by the perturb software behavior of the attacked program by the malware.

  Correct
  Correct.

3. What
are five steps in a typical Anomaly based
Malware Detection Methodology? Hint. Presentation at 7:30 point. **1 / 1 point**

- ( ) 1. Collect data and label them. 2. build model (unsupervised) 3, select features. 4. extract features. 5. Verify detection results/fine tune the model.
- (●) 1. Collect data and label them. 2. select features. 3. extract features. 4. build model (unsupervised) 5. Verify detection results/fine tune the model.

  Correct
  Correct.

4. The
Collecting infrastructure includes the host system be protected and the anomaly
based IDS training/Detection System. In
the host system, they develop kernel driver to collect hardware performance counters
(HPC) and send the measurement values to the training/detection system. Presentation at 8 min point. **1 / 1 point**

Select all that apply.

- [x] By labeling measured data during malware execution period and during clean period, we are able to enable feature selection

  Correct
  Correct.

- [x] By labeling measured data during malware execution period and during clean period, we are able to enable testing and evaluation

  Correct
  Correct.

- [ ] By labeling measured data during malware execution period and during clean period, we are able to detect malware attacks.

5. What
software they use to generate exploit sample. **1 / 1 point**

- ( ) Snort
- (●) Metaploit
- ( ) Stachdral
- ( ) Kali

  Correct
  Correct.

6. What are the challenges in feature selection? **1 / 1 point**

Select all that apply.

- [x] Perturbations
  caused by malware are small in Store, MISP-BR(mispredicted branch instruction),
  MISP_RET (mispredicted near return instruction).

  Correct
  Correct.

- [x] Pick
  events with high Fisher Score (F-Score).

  Correct
  Correct.

7. What is the techniques/tools used to build their Model. **1 / 1 point**

Select all that apply.

- [x] They modify libsvm iibrary developed at National Taiwan University to make their model tunable.

  Correct
  Correct.

- [x] One-class
  Support Vector Machines

  Correct
  Correct.

8. What are two metrics they used to evaluate the results? **1 / 1 point**

Select all that apply.

- [x] Area
  Under Curve (AUC) score

  Correct
  Correct.

- [x] Receiving
  Operating Characteristic (ROC) curve

  Correct
  Correct

- [ ] Ratio between false positive and false negative.